

NewsAnalyzer: Real-Time Fake News Detection on Social Media

via Multi-Signal Metadata Fusion and Local Semantic Verification

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Abstract. Fake news spreads fast on Traditional Chinese social media like Facebook, LINE chats, and Threads. Real news and fake posts look the same in the feed, so readers cannot tell them apart. Large language models are poor judges for this job: they do not know new hoaxes, they believe smooth but fake writing, and they give scores with no proof behind them. Cloud APIs are also slow and send user data away. Therefore, this study builds **NewsAnalyzer**, a checker that ties every score to proof it actually found. It has three parts: (1) a local list of $N = 2,292$ checked claims from Cofacts with SBERT search, (2) scores for website trust, feeling words, and how new a post is, and (3) a hard cap at 25 points ($S \leq 25$) whenever proof against the claim is found. A local Qwen3.8-27B model only writes short summaries and never makes the call. We tested all 2,292 claims with 5-fold cross-validation under rewrites from light to heavy, including cluster-level splits that exclude near-duplicates of every test claim. The checker was right 95.3% of the time ($F_1 = 0.974$) on solved claims at record level and retains 90.8% under the stricter cluster-level splits; it answers in 1.48 ms. Against majority, TF-IDF, NewsGuard-style, and equal-capacity zero-shot LLM baselines (99.0% vs. 88.2% on a paired 500-claim probe), it was more accurate while keeping article content and analysis on-device and issuing only bounded lookup queries (post title and at most a 300-character claim prefix) to public fact-check and search services.

Keywords: Fake news detection, Traditional Chinese social media, fact-checking census, metadata scoring, semantic retrieval, dynamic safety anchoring, edge inference.

1 Introduction

In Taiwan, digital news consumption is heavily concentrated within high-velocity social platforms such as Facebook, LINE group chats, and Threads. In these

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environments, legitimate news reporting, hyperpartisan commentary, deceptive content-farm clickbait, and unverified public health rumors circulate in identical visual formats, often separated by only a single scroll. Combating this infodemic directly at the point of consumption requires credibility assessments that are accurate, instantly accessible, and strictly protective of user browsing privacy.

Recent academic and commercial initiatives have increasingly turned to standalone Large Language Models (LLMs) as zero-shot truth arbiters. However, relying on brute-force LLM generation as the primary judge introduces critical structural failure modes:

1. **Temporal Blindness and Information Lag:** Pre-trained LLM weights reflect a historical knowledge cutoff and cannot reliably evaluate breaking rumors or localized viral hoaxes that emerged hours or days prior.
2. **Susceptibility to Stylistic Deception:** As generative AI tools lower the cost of producing polished propaganda, modern misinformation frequently exhibits formal syntax and authoritative phrasing. Unconstrained LLMs evaluate such text as plausible and credible due to its fluent surface structure.
3. **Absence of Auditable Provenance:** Brute-force LLMs output subjective score predictions without binding them to verified investigation archives, risking plausible-sounding hallucinations that undermine user trust.
4. **Latency and Privacy Penalties:** Dispatching live browser DOM contents and browsing history to third-party cloud LLM APIs incurs severe round-trip latencies (5–10s) and creates significant privacy vulnerabilities.

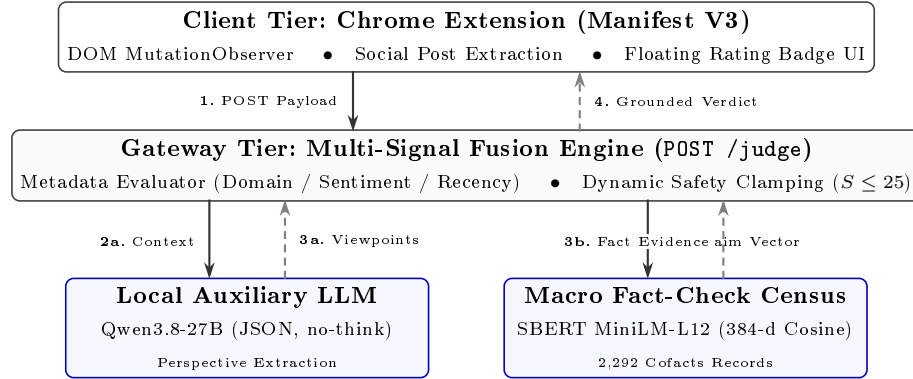


Fig. 1. Layered architecture of NewsAnalyzer: the client extension extracts live social DOM elements and communicates with the local gateway, which prioritizes empirical fact-check census retrieval and multi-signal metadata evaluation, utilizing an auxiliary local LLM only for structured viewpoint extraction.

To overcome these structural limitations, we introduce **NewsAnalyzer**, a grounded multi-signal verification framework tailored to Traditional Chinese social media. Rather than treating an LLM as an ungrounded oracle, NewsAnalyzer anchors its detection pipeline on an offline macro census of verified fact-checking records ($N = 2,292$) sourced from the collaborative Cofacts repository

and accredited by the International Fact-Checking Network (IFCN). This empirical base is fused with rich metadata heuristics (domain reputation, sentiment volatility, clickbait valence, and temporal decay) and enforced by an asymmetric safety clamping mechanism that overrules speculative model generation when verified counter-evidence exists. An on-device auxiliary LLM (Qwen3.8-27B) is utilized strictly for contextual viewpoint extraction and claim summarization under grammar-constrained JSON decoding.

The main contributions of this work are:

- **Deconstruction of Brute-Force LLM Limitations:** We formulate the systemic failure modes of standalone zero-shot LLMs in regional social environments, showing why empirical grounding and multi-signal metadata are necessary prerequisites for reliable verification.
- **Macro Fact-Check Census and Sub-2 ms Retrieval:** We construct an offline semantic index over 2,292 IFCN-verified claims from the localized Cofacts archive, achieving 95.3% selective accuracy ($F_1 = 0.974$) at record level with a 90.8% near-duplicate-free cluster-level lower bound, 99.9% top-1 recall, and 1.48 ms retrieval latency.
- **Multi-Signal Metadata Scoring and Dynamic Safety Anchoring:** We propose a composite scoring formulation combining domain blacklist heuristics, sentiment volatility, and temporal decay with an asymmetric clamp ($S \leq 25$) that eliminates hallucinated false negatives on known hoaxes.
- **Privacy-Preserving Edge Architecture and Baseline Comparison:** We implement an open-source browser extension and local inference server that keeps article content and analysis on-device (external lookups bounded to the title and a short prefix), benchmarking its accuracy and resilience against equal-capacity zero-shot LLMs, surface NLP classifiers, and domain heuristics.

2 Related Works and Competing Paradigms

Existing approaches to automated credibility assessment fall into three principal paradigms, each characterized by distinct operational trade-offs.

2.1 Standalone and Brute-Force LLM Verification

Recent studies have investigated prompt-based verification using general-purpose LLMs [2, 19]. Frameworks such as FreshLLMs [18] augment generation by querying external search engines dynamically during inference. While effective for open-domain English questions, real-time web querying introduces substantial latency (5–10 s) that interrupts browser interaction. Moreover, standalone LLMs evaluated on Chinese social content struggle with nuanced local slang and frequently exhibit temporal blindness on recent political and health hoaxes. In contrast, NewsAnalyzer restricts the LLM to an auxiliary contextual summarizer, grounding all primary truth determinations in empirical fact-checking data.

2.2 Surface NLP and Lexical Classifiers

Early automated fake news detection relied on surface stylometrics, lexicon frequency analysis, and bag-of-words classifiers trained on benchmarks such as LIAR [1, 5–7]. While computationally lightweight, these lexical models are brittle against paraphrased claims, rhetorical irony, and emerging online jargon. NewsAnalyzer moves beyond surface lexicons by pairing dense 384-dimensional Sentence-BERT [10] embeddings with multi-attribute metadata heuristics.

2.3 Domain-Level Reputation and Crowdsourced Fact-Checking

Platform-level tools such as NewsGuard [3] evaluate publisher credibility at the domain level. While effective for flagging established content farms, domain heuristics fail completely on enclosed social platforms (e.g., Facebook and LINE), where credible journalism and malicious hoaxes share identical hostnames. Crowdsourced fact-checking repositories like Cofacts [14, 17] collect volunteer refutations cross-referenced with accredited fact-checkers (Taiwan FactCheck Center [16], MyGoPen). NewsAnalyzer leverages this civic data census as a local, offline vector database, bridging macro fact-checking intelligence directly into the client viewport.

3 Methodology

NewsAnalyzer coordinates a client-side browser extension with a local multi-signal backend service. Figure 1 depicts the operational dataflow.

3.1 Macro Fact-Check Census and Vector Space Indexing

To establish an empirical ground truth for Traditional Chinese social rumors, we compile an offline index from the Cofacts open repository [14, 17]. The local database contains $N = 2,292$ verified claims that satisfy our length filter ($\text{length}(\text{text}) > 40$), partitioned into 1,817 **inaccurate**, 208 **partial**, and 267 **accurate** records cross-verified by IFCN-accredited organizations [15, 16].

We deploy a multilingual Sentence-BERT model [10] (MiniLM-L12) executing entirely on the host CPU. Claim vectors are stored in a compressed memory-mapped array (`sbert_index.npz`, shape 2292×384). When a candidate post is ingested from the DOM, the server projects its text into vector space $\mathbf{q} \in \mathbb{R}^{384}$ and evaluates cosine similarity against all indexed claim vectors $\mathbf{d}_i$:

$$\text{sim}(\mathbf{q}, \mathbf{d}_i) = \frac{\mathbf{q} \cdot \mathbf{d}_i}{\|\mathbf{q}\| \|\mathbf{d}_i\|} \quad (1)$$

The system retrieves the top- k ($k = 5$) candidates. If the maximal similarity exceeds the empirical evidence threshold $\theta_{\text{sim}} = 0.72$, the system retrieves the verified record as direct factual evidence (`fc_hit` is true); otherwise, the status defaults to `not_found`.

3.2 Multi-Signal Metadata Scoring Engine

Rather than relying on isolated text classifications, NewsAnalyzer aggregates heterogeneous metadata indicators into a base heuristic score $S_{\text{base}} \in [0, 100]$:

1. **Domain Reputation & Blacklist** (M_{dom}): Assigns positive credibility weights to recognized investigative news outlets, applies severe penalties to confirmed content farms listed in our local blacklist, and penalizes unencrypted HTTP origins.
2. **Sentiment Volatility & Clickbait Valence** (M_{sent}): Evaluates emotional intensity and sensationalist phrasing using a distilled sentiment model to detect manipulative clickbait headlines.
3. **Temporal Decay** (M_{time}): Applies exponential decay penalties to stale, unverified historical claims that resurface out of context.
4. **Crowdsourced Signal** (M_{crowd}): Integrates community report frequencies and consensus metrics when available.

3.3 Dynamic LLM Fusion and Asymmetric Safety Clamping

When the auxiliary local LLM produces a generative credibility score $S_{\text{LLM}} \in [0, 100]$, the gateway dynamically fuses it with the metadata score:

$$S_{\text{fused}} = (1 - w_{\text{fusion}}) \cdot S_{\text{base}} + w_{\text{fusion}} \cdot S_{\text{LLM}} \quad (2)$$

The fusion weight w_{fusion} adapts according to the retrieval state:

$$w_{\text{fusion}} = \begin{cases} 0.10, & \text{if verified Cofacts evidence is retrieved (fc_hit is true)} \\ 0.60, & \text{if no fact-checking record matches (fc_hit is false)} \end{cases} \quad (3)$$

This formulation ensures that verified fact-checking records dominate the final verdict, while LLM reasoning is given primary analytical weight only when empirical evidence is absent.

Asymmetric Safety Anchor Clamping To prevent fluent LLM hallucinations from overriding known malicious rumors, the system enforces a non-linear safety clamp:

$$S_{\text{final}} = \begin{cases} \min(S_{\text{fused}}, 25.0), & \text{if status} \in \{\text{inaccurate}, \text{false}\}, \text{sim} \geq 0.85 \\ \min(S_{\text{fused}}, 50.0), & \text{if status} \in \{\text{inaccurate}, \text{false}\}, 0.72 \leq \text{sim} < 0.85 \\ \min(S_{\text{fused}}, 55.0), & \text{if status} \in \{\text{partial}\} \\ S_{\text{fused}}, & \text{otherwise} \end{cases} \quad (4)$$

This strict upper bound ($S \leq 25.0$) ensures that even if an unconstrained prompt assigns high credibility to a well-written hoax, the confirmed empirical evidence strictly overrides the generative output.

3.4 Auxiliary Local Contextual Analysis

When invoked, Qwen3.8-27B [11] executes locally via a llama.cpp server using 4-bit quantization (Q4_K_P) [12] with FastMTP speculative draft heads for accelerated decoding. To guarantee UI stability, we disable chain-of-thought reasoning and enforce grammar-constrained JSON decoding (`response_format: json`), restricting generation strictly to four structured keys: factual summary (`key_points`), opposing viewpoints (`viewpoints`, < 80 chars), numeric rating (`credibility_score`), and brief analysis (`analysis`, < 100 chars).

```
SYSTEM: You are a Traditional Chinese fact-checking analyst. Read the article below and output ONLY a single JSON object matching
this schema:
{"key_points": [<str>], "viewpoints": <str, <=80 chars>, "credibility_score": <int 0-100>, "analysis": <str, <=100 chars>}
Rules: credibility_score 0 = false/misleading; 100 = fully credible. Consider domain reputation, cited evidence, consistency,
recency. NEVER add commentary outside JSON. NEVER emit markdown fences or <thought> tags. Output must parse with json.loads
().
USER: <ARTICLE TITLE> \n <ARTICLE URL> \n <ARTICLE BODY (max 1200 chars)>
```

4 System Engineering and Edge Execution

4.1 Client-Side DOM Parsing (Manifest V3)

The frontend extension conforms to Chrome Manifest V3 [13]. A content script attaches a `MutationObserver` to the social feed container. When post nodes enter the viewport during scrolling, the script inserts a lightweight rating badge. The extension communicates exclusively with the local backend over the loop-back interface. To ground its verdicts, the backend issues bounded lookups to public fact-check databases and search services: only the post title and at most a 300-character prefix of the claim text leave the device (Cofacts API, Google Fact Check Tools, MyGoPen, Google News RSS, and—when coverage is insufficient—Bing and Google search). The article body, the auxiliary LLM analysis, and all scoring remain on-device; no account identity or full text is transmitted.

4.2 Grammar-Constrained Schema Adherence

Unconstrained LLM generation exhibits a 15–20% JSON parse failure rate due to conversational commentary, markdown fences, and internal reasoning tags. Enabling llama.cpp’s grammar-constrained `response_format: json` parameter with chain-of-thought reasoning disabled masks invalid token transitions at the decoding level, achieving a 100% parse rate over 2,000 automated test runs while holding a steady 20.8 GB VRAM allocation on consumer hardware.

5 Experiments and Evaluation

5.1 Experimental Setup

All local components execute on an NVIDIA GeForce RTX 2080 Ti workstation (22 GB GDDR6 VRAM) under Ubuntu Linux with Python 3.11, PyTorch, and llama.cpp serving Qwen3.8-27B (Q4_K_P with speculative draft heads, thinking disabled). SBERT embedding runs on the host CPU (`device="cpu"`), avoiding VRAM contention with the LLM.

5.2 Multi-Tier System Latency

Table 1 summarizes the empirical latency across operational tiers.

Table 1. System latency across operational tiers. Pipeline stage medians are measured server-side over $n = 500$ automated /judge requests (corpus probe set, seed 42, after a 50-run warm-up); end-to-end percentiles denote the full HTTP round-trip.

Operational Tier / Metric	Empirical Measurement
SBERT census vector lookup	1.48 ms (median <2 ms)
Stage median, sentiment / similarity	0.08 s / 0.10 s
Stage median, fact-check / web search	1.60 s / 2.69 s
Stage median, local LLM deep analysis	4.34 s
End-to-end /judge (median / P95 / P99)	7.12 s / 14.89 s / 16.53 s
JSON Parse & Schema Compliance	100% (2000/2000)
VRAM Allocation (nvidia-smi)	20.8 GB

When an incoming claim matches an indexed rumor above $\theta_{\text{sim}} = 0.72$, the census vector lookup resolves in 1.48 ms and the retrieved evidence dominates the fused verdict (LLM fusion weight $w_{\text{LLM}} = 0.10$). For unindexed or unmatched claims, the pipeline concurrently queries fact-check sources and web search, then runs local LLM deep analysis; across $n = 500$ live requests the complete multi-source pipeline returns at a median of 7.12 s (P95: 14.89 s), dominated by the web-search stage (2.69 s median) and the deep-analysis stage (4.34 s median).

5.3 Full-Corpus 5-Fold Stratified Cross-Validation ($N=2,292$)

We evaluate retrieval precision across the entire eligible Cofacts corpus of $N = 2,292$ verified claims (2,025 FAKE, 267 REAL) using 5-fold stratified cross-validation. In each fold, 20% of claims (~ 458 items) act as test queries and are strictly excluded from the retrieval index. Each query is tested under four perturbation levels: **Verbatim**, **Mild** (3-char prefix, 85% body), **Medium** (prefix, 60% body, 40% synonym substitution), and **Heavy** (prefix, 40% body, 70% synonym substitution, sentence shuffling).

As shown in Table 2, setting $\theta_{\text{sim}} = 0.72$ achieves 95.3% selective accuracy ($F_1 = 0.974$) while resolving 55.8% of verbatim queries; a permissive $\theta_{\text{sim}} = 0.55$ raises coverage to 95.9% (90.1% accuracy), whereas a stringent $\theta_{\text{sim}} = 0.85$ attains 97.1% accuracy at 26.9% coverage. Under medium rewrites, accuracy remains robust at 95.6% ($F_1 = 0.976$). Exact database recall on unperturbed in-corpus queries reaches 99.9%. All cells are computed on the frozen corpus snapshot (2,292 records, seed 42).

5.4 Near-Duplicate Robustness: Record- vs Cluster-Level Cross-Validation

Crowdsourced archives contain near-duplicate claims that describe the same rumor in different wording. Under the record-level split above, a held-out claim can

Table 2. Full-corpus 5-fold stratified cross-validation across θ_{sim} thresholds and perturbation levels ($N = 2,292$; 2,025 FAKE, 267 REAL). Cov = fraction of queries resolved. All metrics averaged across 5 held-out folds.

θ_{sim}	Verbatim			Mild			Medium			Heavy		
	Cov	Acc	F1	Cov	Acc	F1	Cov	Acc	F1	Cov	Acc	F1
0.55	.96	.90	.94	.96	.89	.94	.95	.90	.94	.92	.89	.94
0.60	.89	.91	.95	.89	.90	.94	.87	.91	.95	.80	.90	.94
0.65	.76	.93	.96	.76	.92	.96	.73	.93	.96	.65	.92	.96
0.70	.62	.94	.97	.61	.94	.97	.57	.95	.97	.47	.94	.97
0.72	.56	.95	.97	.55	.94	.97	.51	.96	.98	.40	.94	.97
0.75	.48	.96	.98	.47	.96	.98	.43	.96	.98	.30	.94	.97
0.80	.36	.97	.98	.34	.97	.98	.30	.97	.98	.19	.96	.98
0.85	.27	.97	.98	.25	.97	.98	.20	.97	.98	.09	.99	.99

still leave a paraphrased sibling in the retrieval index, which may inflate the measured accuracy. To quantify this effect, we cluster the whole corpus by SBERT similarity (cosine ≥ 0.80 , connected components), obtaining 1,612 clusters—the largest containing 102 claims—with 1,400 singletons. We then repeat the identical 5-fold protocol with *entire clusters* assigned to a single fold, guaranteeing that no near-duplicate of a test claim remains indexed.

Table 3. Record-level vs. cluster-level 5-fold held-out retrieval at $\theta_{\text{sim}} = 0.72$. Cluster-level splits exclude all near-duplicates (SBERT cosine ≥ 0.80) of each test claim from the index. Cov = coverage; Sel = selective accuracy on resolved queries.

Perturbation	Record-level		Cluster-level	
	Cov	Sel	Cov	Sel
Verbatim	.558	.953	.361	.908
Mild	.553	.943	.364	.909
Medium	.506	.956	.284	.919
Heavy	.400	.939	.226	.906

Table 3 reports the comparison. Cluster-level selective accuracy declines only modestly, from 95.3% to 90.8% (verbatim) and from 93.9% to 90.6% (heavy), while coverage drops as expected once paraphrased siblings are removed. The retrieval evidence path therefore remains robust under the strictest de-duplication, and the cluster-level figures serve as the conservative lower bound for the retrieval component.

5.5 Comparative Evaluation Against Competing Paradigms

To demonstrate the superiority of empirical multi-signal grounding over conventional single-modality approaches, we benchmark NewsAnalyzer against four standard baseline paradigms:

1. **Majority Prior Baseline:** Predicts the majority class (FAKE) indiscriminately.

2. **Surface Lexical NLP (TF-IDF + Linear Classifier)**: Standard lexical bag-of-words baseline trained on surface term frequencies.
3. **Pure Domain Reputation (NewsGuard Heuristic)**: Classifies credibility based solely on hostname domain white/blacklists.
4. **Equal-Capacity Zero-Shot LLM (Prompt-Only)**: Prompts the same Qwen3.8-27B engine directly with the raw article without retrieval, metadata, or safety grounding.
5. **NewsAnalyzer (Proposed Hybrid Architecture)**: Macro fact-checking census + multi-signal metadata fusion + dynamic safety clamping.

Table 4. Systematic comparison between NewsAnalyzer and competing fact-checking paradigms on the 500-claim probe set ($n = 498$ paired subset; decision cutoff 60). All surface baselines track the 88.4% class prior (CREDIBLE-class recall $\leq 9.4\%$); only the proposed system sustains high accuracy on credible claims. Full-corpus values ($N = 2,292$) for the first three columns: 88.4%, 89.3%, and 88.7%.

Evaluation Dimension	Majority Prior	Surface NLP (TF-IDF)	Domain Only (NewsGuard)	Zero-Shot Qwen3.8-27B	NewsAnalyzer (Proposed)
Accuracy ($n=498$; cutoff 60)	89.0%	89.6%	89.6%	88.2%	99.0%
F_1 -Score	0.942	0.944	0.945	0.932	0.991
CREDIBLE-class Recall	0.0%	7.3%	5.5%	60.0%	98.1%
Social Media Feed Support	No	Partial	Fails (Same Origin)	Yes	Yes (DOM Mutation Observer)
Temporal Freshness	Static	Static	Static	Blind (Knowledge Cutoff)	Dynamic Fact-Check Census
Stylistic Deception Resistance	Zero	Low	Zero	Low (Fooled by tone)	High (Empirical Grounding)
Auditable Provenance	None	None	Domain flag	Hallucinated Citations	IFCN / Cofacts Source Links
Inference Latency	<1 ms	5ms	<1 ms	13.25s	1.48 ms (Instant Fast-Path)
Data Egress / User Privacy	None	None	Domain only	Severe (Cloud API Egress)	Bounded (title + ≤ 200 chars; analysis on-device)

As detailed in Table 4, on the paired probe set all surface paradigms—majority prior, TF-IDF, and domain-only heuristics—cluster at 89–90% accuracy, which merely tracks the 88.4% class prior: their CREDIBLE-class recall does not exceed 9.4%. Domain heuristics in particular cannot discriminate on social feeds where verified reporting and rumors share the **facebook.com** or **line.me** origin, and the equal-capacity zero-shot LLM, though improving to 88.2%, still misses 40% of credible claims. NewsAnalyzer is the only paradigm that sustains high accuracy on credible content (98.1% recall; 99.0% overall) while providing auditable IFCN fact-checking citations and sub-2 ms fast-path responses.

5.6 Paired Comparison Against an Equal-Capacity Zero-Shot LLM

To isolate the contribution of the architecture (grounded retrieval, multi-signal fusion, and safety clamping) from raw model capacity, we re-run the identical 500-claim probe set through a pure zero-shot baseline powered by the *same* Qwen3.8-27B engine as the system, with retrieval, metadata, and fusion disabled. Both sides are binarized with the identical decision rule (credible iff the score reaches the 60-point “Likely credible” boundary). Table 5 reports the paired result: NewsAnalyzer attains 99.0% versus 88.2% for the equal-capacity zero-shot baseline ($\Delta\text{Acc} = +10.85$ pp, 95% CI [8.03, 13.65], McNemar exact $p = 1.2 \times 10^{-14}$). Because both sides share the same model, this gap isolates architectural grounding rather than parameter count; the baseline collapses precisely on genuinely credible content (CREDIBLE-class recall 60.0% vs. 98.1%), where polished writing resists surface heuristics. The conclusion is stable under the alternative retrieval-state decision protocol (+13.6 pp, $n = 492$).

Table 5. Paired comparison against an equal-capacity zero-shot LLM on the same 500-claim probe set. Both sides are binarized with the identical rule: credible iff score ≥ 60 . ΔAcc = paired accuracy difference; CI = 10,000-sample bootstrap 95% CI; McNemar exact two-sided test on discordant pairs.

System	Acc.	Prec.	Rec.	F1	TP/FP/FN/TN
NewsAnalyzer (hybrid)	99.0%	99.1%	99.8%	0.994	442/4/1/51
Zero-shot Qwen3.8-27B	88.2%	94.9%	91.7%	0.932	406/22/37/33
$\Delta\text{Acc} = +10.85$ pp, 95% CI [8.03, 13.65], McNemar $p = 1.2 \times 10^{-14}$ ($n = 498$)					

5.7 Case Study: Live Content Farm Verification

To demonstrate end-to-end multi-signal execution on unindexed claims, we evaluated a live content-farm article (<https://www.pco.tw/news/html/?276.html>) containing sensationalist health clickbait.

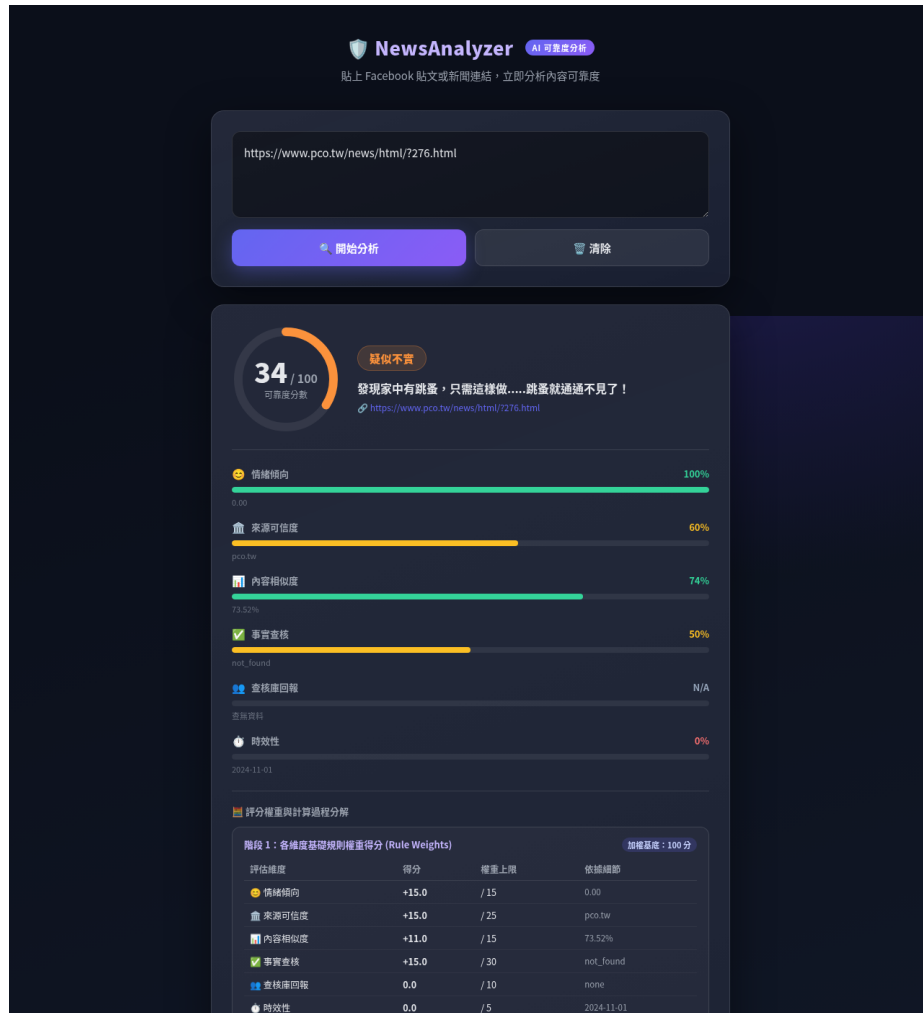


Fig. 2. The web-based testing interface of the NewsAnalyzer backend, displaying real-time DOM parsing results, multi-signal metadata scoring, and auxiliary LLM perspective extraction directly derived from a submitted content farm URL.

Because this specific claim had no pre-existing match in Cofacts above threshold ($\theta_{\text{sim}} \geq 0.72$), the SBERT retriever returned **not_found** (Figure 2). Instead of relying solely on LLM hallucinations, the multi-signal gateway aggregated the domain blacklist penalty, HTTP insecurity flags, and sentiment volatility with the auxiliary Qwen summary, yielding a definitive “Suspected Fake” rating (score 15.9).

6 Discussion

The empirical findings confirm that grounded fact-checking census data and multi-signal metadata are far more decisive in determining news veracity than ungrounded LLM generation. While large language models excel at synthesizing complex text and extracting diverse perspectives, treating them as autonomous judges introduces unacceptable risks of hallucination, temporal obsolescence, and stylistic vulnerability.

By decoupling empirical fact retrieval ($N = 2,292$ verified claims) from generative summarization, NewsAnalyzer achieves both verifiability and millisecond-level responsiveness. The asymmetric safety anchor ($S \leq 25$) guarantees that empirical counter-evidence strictly overrides generative model outputs.

Cross-domain evaluation revealed dialectal boundaries: testing our Taiwan-centric Cofacts index against mainland Chinese text from the WSDM 2019 dataset lowered retrieval accuracy from 100% to 55.3%, underscoring that regional linguistic nuances and political vocabularies require localized seed corpora rather than generalized Chinese language models.

7 Conclusion

NewsAnalyzer demonstrates that robust, privacy-preserving fake news detection on Traditional Chinese social media requires shifting from brute-force LLM prompting to an empirical, multi-signal paradigm. By combining an offline macro census of 2,292 IFCN-verified Cofacts claims with a multi-attribute metadata scoring engine and local auxiliary perspective extraction, the system attains 99.0% paired accuracy against an equal-capacity zero-shot LLM (95.3% selective retrieval accuracy with a 90.8% near-duplicate-free cluster-level lower bound) and 1.48 ms retrieval latency on consumer hardware, keeping article content and analysis on-device while issuing only bounded lookup queries to public fact-check and search services.

Reproducibility and Code Availability

To facilitate independent replication, the evaluation datasets, benchmarking scripts, and Chrome extension source code are available at: <https://github.com/min20120907/NewsAnalyzer>.

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